

If $Z \sim N(0, 1)$ then $E(Z^3) = 0, E(Z^4) = 3$. Let W_t be a Wiener process.

Financial Mathematics (MA30247/MA60225)		Set-1	
Name:		Class Test-1	
Roll No.		Marks:20	Time: 45 mins
Q.N.	Question	Answer	Marks
1.	Compute the distribution of stochastic integral $\int_0^1 t dW_t$.	$N(0, 1/3)$	2
2.	Compute $E(W_2^2 W_3^2)$	14	2
3.	$X(t) = e^{-t}W(e^{2t})$ is a Gaussian process. (Yes/No)	Yes	2
4.	Let $\{W_t\}_{t \geq 0}$ be a Wiener process. Then $4W_1 + 6W_3 \sim$	$N(0, 172)$	2
5.	Is the process $Y_t := e^{t/2} \sin W_t$ a martingale? (Yes/No)	Yes	2
6.	Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion (GBM) with drift parameter $\mu = 0.01$ and volatility parameter $\sigma = 0.3$. If $Y(0) = 5$. Then, find the value of $E[Y(10)]$.	$5e^{0.55}$ or $5e^{0.1}$	2
7.	Weiner process is a Gaussian process. (T/F)	T	2
8.	Which of the following statements is TRUE? (a) Every stochastic process is a martingale. (b) The mean of a martingale process is time varying. (c) The mean of a martingale process is constant with respect to time. (d) The mean of a martingale process need not exist.	(c)	2
9.	Let $Y_t := e^{W_t}$. Then $Cov(Y_1, Y_2) =$	$(e - 1)e^{3/2}$	2
10.	A person is playing a game in which he/she wins or loses one rupee in each round of play, with equal chances and independently of the past outcomes. If he/she starts playing with the firm determination that he/she will stop playing when either he/she won 10 rupees or she lost 5 rupees. (a) What is the probability that he/she will be winning when he/she stops playing further. (b) What is the expected number of his/her playing rounds before he/she will stop playing further.	1/3, 50	2

If $Z \sim N(0, 1)$ then $E(Z^3) = 0, E(Z^4) = 3$. Let W_t be a Wiener process.

Financial Mathematics (MA30247/MA60225)		Code: Set-2	
Name:		Class Test-1	
Roll No.		Marks:20	Time: 45 mins
Q.N.	Question	Answer	Marks
1.	Compute the distribution of stochastic integral $\int_0^1 t^2 dW_t$.	$N(0, 1/5)$	2
2.	Compute $E(W_1^2 W_3^2)$	5	2
3.	$X(t) = e^{-2t}W(e^{2t})$ is a Gaussian process. (Yes/No)	Yes	2
4.	Let $\{W_t\}_{t \geq 0}$ be a Wiener process. Then $3W_1 + 6W_3 \sim$	$N(0, 153)$	2
5.	Is the process $Y_t := e^t \cos W_t$ a martingale? (Yes/No)	No	2
6.	Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion (GBM) with drift parameter $\mu = 0.02$ and volatility parameter $\sigma = 0.4$. If $Y(0) = 10$. Then, find the value of $E[Y(10)]$.	$10e$ or $10e^{0.2}$	2
7.	Weiner process is not a Gaussian process. (T/F)	F	2
8.	Which of the following statements is TRUE? (a) The mean of a martingale process is constant with respect to time. (b) The mean of a martingale process need not exist. (c) Every stochastic process is a martingale. (d) The mean of a martingale process is time varying.	(a)	2
9.	Let $Y_t := e^{W_t}$. Then $Cov(Y_2, Y_3) =$	$e^{5/2}(e^2 - 1)$	2
10.	A person is playing a game in which he/she wins or loses one rupee in each round of play, with equal chances and independently of the past outcomes. If he/she starts playing with the firm determination that he/she will stop playing when either he/she won 20 rupees or she lost 5 rupees. (a) What is the probability that he/she will be winning when he/she stops playing further. (b) What is the expected number of his/her playing rounds before he/she will stop playing further.	$1/5, 100$	2

Financial Mathematics (MA30247/MA60225)		Code: Set-3	
Name:		Class Test-1	
Roll No.		Marks:20	Time: 45 mins
Q.N.	Question	Answer	Marks
1.	Compute the distribution of stochastic integral $\int_0^1 t^3 dW_t$.	$N(0, 1/7)$	2
2.	Compute $E(W_2^2 W_4^2)$	16	2
3.	$X(t) = e^{-t}W(e^t)$ is a Gaussian process. (Yes/No)	Yes	2
4.	Let $\{W_t\}_{t \geq 0}$ be a Wiener process. Then $4W_1 + 5W_3 \sim$	$N(0, 131)$	2
5.	Is the process $Y_t := e^t \sin W_t$ a martingale? (Yes/No)	No	2
6.	Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion (GBM) with drift parameter $\mu = 0.03$ and volatility parameter $\sigma = 0.2$. If $Y(0) = 5$. Then, find the value of $E[Y(10)]$.	$5e^{0.5}$ or $5e^{0.3}$	2
7.	Weiner process is a Gaussian process. (T/F)	T	2
8.	Which of the following statements is TRUE? (a) Every stochastic process is a martingale. (b) The mean of a martingale process need not exist. (c) The mean of a martingale process is time varying. (d) The mean of a martingale process is constant with respect to time.	(d)	2
9.	Let $Y_t := e^{W_t}$. Then $Cov(Y_2, Y_4) =$	$e^3(e^2 - 1) \approx 948.43$	2
10.	A person is playing a game in which he/she wins or loses one rupee in each round of play, with equal chances and independently of the past events. If he/she starts playing with the firm determination that he/she will stop playing when either he/she won 30 rupees or she lost 10 rupees. (a) What is the probability that he/she will be winning when he/she stops playing further. (b) What is the expected number of his/her playing rounds before he/she will stop playing further.	$1/4, 300$	2

Q.N.	Question	Answer	Marks
1.	Compute the distribution of stochastic integral $\int_0^1 t^4 dW_t$.	$N(0, 1/9)$	2
2.	Compute $E(W_1^2 W_4^2)$	6	2
3.	$X(t) = e^{-3t} W(e^{2t})$ is a Gaussian process. (Yes/No)	Yes	2
4.	Let $\{W_t\}_{t \geq 0}$ be a Wiener process. Then $4W_1 + 7W_3 \sim$	$N(0, 219)$	2
5.	Is the process $Y_t := e^{t/2} \cos W_t$ a martingale? (Yes/No)	Yes	2
6.	Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion (GBM) with drift parameter $\mu = 0.01$ and volatility parameter $\sigma = 0.2$. If $Y(0) = 5$. Then, find the value of $E[Y(10)]$.	$5e^{0.3}$ or $5e^{0.1}$	2
7.	Weiner process is not a Gaussian process. (T/F)	F	2
8.	Which of the following statements is TRUE? (a) The mean of a martingale process is time varying. (b) The mean of a martingale process is constant with respect to time. (c) Every stochastic process is a martingale. (d) The mean of a martingale process need not exist.	(b)	2
9.	Let $Y_t := e^{W_t}$. Then $Cov(Y_1, Y_4) =$	$e^{5/2}(e - 1) \approx 20.933$	2
10.	A person is playing a game in which he/she wins or loses one rupee in each round of play, with equal chances and independently of the past events. If he/she starts playing with the firm determination that he/she will stop playing when either he/she won 6 rupees or she lost 5 rupees. (a) What is the probability that he/she will be winning when he/she stops playing further. (b) What is the expected number of his/her playing rounds before he/she will stop playing further.	5/11, 30	2

1 CT-1 Solution, Financial Math 2026 : Set-1

Question 1:

Compute the distribution of stochastic integral $\int_0^1 t dW_t$.

Answer: $N(0, 1/3)$

Solution

$$\mathbb{E} \left[\int_0^1 t dW_t \right] = 0$$

According to Itô Isometry, the variance of the stochastic integral is computed by integrating the square of the deterministic integrand:

$$\text{Var}(X) = \mathbb{E} \left[\left(\int_0^1 t dW_t \right)^2 \right] = \int_0^1 t^2 dt = \frac{1}{3}$$

Therefore $\int_0^1 t dW_t \sim N(0, 1/3)$

Answer: $N(0, 1/3)$

Question 2:

Compute $E(W_2^2 W_3^2)$

Answer: 14

Solution By the definition of a standard Brownian motion, the process has independent increments. We can decompose W_3 into the sum of W_2 and the independent increment $(W_3 - W_2)$.

Let $Z = W_3 - W_2$. Then:

- $W_2 \sim N(0, 2)$, so $E(W_2^2) = 2$
- $Z \sim N(0, 1)$, so $E(Z^2) = 1$ and $E(Z) = 0$
- W_2 and Z are independent

Expressing W_3 as $W_2 + Z$, we expand the expectation:

$$\begin{aligned} E(W_2^2 W_3^2) &= E[W_2^2 (W_2 + Z)^2] \\ &= E[W_2^2 (W_2^2 + 2W_2 Z + Z^2)] \\ &= E(W_2^4) + 2E(W_2^3 Z) + E(W_2^2 Z^2) \end{aligned}$$

We now evaluate each term individually using the properties of Gaussian distributions and independence:

1. **First Term:** For a zero-mean normal random variable, the fourth moment is given by $3\sigma^4$:

$$E(W_2^4) = 3 \times (\text{Var}(W_2))^2 = 3 \times 2^2 = 12$$

2. **Second Term:** Due to the independence of W_2 and Z , and the fact that $E(Z) = 0$:

$$2E(W_2^3 Z) = 2E(W_2^3)E(Z) = 2E(W_2^3) \times 0 = 0$$

3. **Third Term:** Due to the independence of W_2^2 and Z^2 :

$$E(W_2^2 Z^2) = E(W_2^2)E(Z^2) = 2 \times 1 = 2$$

Summing these results together gives:

$$E(W_2^2 W_3^2) = 12 + 0 + 2 = 14$$

Final Answer: The value of $E(W_2^2 W_3^2)$ is 14.

Question 3:

$X(t) = e^{-t}W(e^{2t})$ is a Gaussian process. (Yes/No)

Answer: Yes

Solution A stochastic process $\{X(t), t \geq 0\}$ is a **Gaussian process** if for any finite set of time instances $t_1, t_2, \dots, t_n$, the joint distribution of the random variables $X(t_1), X(t_2), \dots, X(t_n)$ is a multivariate Gaussian distribution.

1. **Base Process:** Let $W(t)$ be a standard Brownian motion (Wiener process). By definition, $W(t)$ is a Gaussian process.
2. **Time Rescaling:** The term $W(e^{2t})$ represents a deterministic time-rescaling of the Brownian motion. Since this transformation only changes the time index linearly/deterministically, any finite collection $(W(e^{2t_1}), \dots, W(e^{2t_n}))$ remains jointly Gaussian.
3. **Linear Scaling:** The factor e^{-t} is a deterministic scaling function. Multiplying a jointly Gaussian vector by deterministic constants is a linear transformation.

Since linear transformations of Gaussian random variables preserve joint Gaussianity, the process $X(t) = e^{-t}W(e^{2t})$ is inherently a Gaussian process.

Answer: Yes

Question 4:

Let $\{W_t\}_{t \geq 0}$ be a Wiener process. Then $4W_1 + 6W_3 \sim$

Answer: $4W_1 + 6W_3 \sim N(0, 172)$

Solution

Let $\{W_t\}_{t \geq 0}$ be a Wiener process. We want to find the distribution of the linear combination $4W_1 + 6W_3$.

Since any linear combination of jointly Gaussian random variables is normally distributed, we know that:

$$4W_1 + 6W_3 \sim N(\mu, \sigma^2)$$

1. Mean (μ)

A Wiener process has a mean of zero for all t , meaning $E[W_t] = 0$. By linearity of expectation:

$$\mu = E[4W_1 + 6W_3] = 4E[W_1] + 6E[W_3] = 4(0) + 6(0) = 0$$

2. Variance (σ^2)

Using the variance identity for a sum of random variables:

$$\text{Var}(4W_1 + 6W_3) = 16\text{Var}(W_1) + 36\text{Var}(W_3) + 48\text{Cov}(W_1, W_3)$$

For a standard Wiener process, we use the properties $\text{Var}(W_t) = t$ and $\text{Cov}(W_s, W_t) = \min(s, t)$:

- $\text{Var}(W_1) = 1$
- $\text{Var}(W_3) = 3$
- $\text{Cov}(W_1, W_3) = \min(1, 3) = 1$

Substituting these values into our variance formula:

$$\sigma^2 = 16(1) + 36(3) + 48(1)$$

$$\sigma^2 = 16 + 108 + 48 = 172$$

Conclusion

$$4W_1 + 6W_3 \sim N(0, 172)$$

Answer: $4W_1 + 6W_3 \sim N(0, 172)$

Question 5:

Is the process $Y_t := e^{t/2} \sin W_t$ a martingale? (Yes/No)

Answer: Yes

Solution To prove that the process $Y_t = e^{t/2} \sin W_t$ is a martingale using the definition of conditional expectation, we must show that for any $0 \leq s \leq t$:

$$\mathbb{E}[Y_t \mid \mathcal{F}_s] = Y_s$$

Step 1: Use Independent Increments

We isolate the new information after time s by rewriting W_t as $(W_t - W_s + W_s)$. Using the trigonometric identity $\sin(A+B) = \sin A \cos B + \cos A \sin B$, we expand the process:

$$Y_t = e^{t/2} \left[\sin(W_t - W_s) \cos W_s + \cos(W_t - W_s) \sin W_s \right]$$

Step 2: Apply Conditional Expectation

Taking the conditional expectation with respect to the filtration $\mathcal{F}_s$:

$$\mathbb{E}[Y_t \mid \mathcal{F}_s] = e^{t/2} \left[\mathbb{E}[\sin(W_t - W_s) \cos W_s \mid \mathcal{F}_s] + \mathbb{E}[\cos(W_t - W_s) \sin W_s \mid \mathcal{F}_s] \right]$$

Because W_s is $\mathcal{F}_s$ -measurable and the increment $(W_t - W_s)$ is independent of $\mathcal{F}_s$, we can extract the measurable components and remove the conditioning from the independent components:

$$\mathbb{E}[Y_t \mid \mathcal{F}_s] = e^{t/2} \left[\cos W_s \cdot \mathbb{E}[\sin(W_t - W_s)] + \sin W_s \cdot \mathbb{E}[\cos(W_t - W_s)] \right]$$

Step 3: Evaluate Expectations Using Normal Distribution

Since the Brownian motion increment follows a normal distribution, $(W_t - W_s) \sim N(0, t - s)$:

1. By symmetry of the normal distribution, the expectation of the odd function is zero:

$$\mathbb{E}[\sin(W_t - W_s)] = 0$$

2. Using the characteristic function of a normal distribution $\mathbb{E}[e^{iuX}] = e^{-\frac{1}{2}u^2\sigma^2}$ evaluated at $u = 1$:

$$\mathbb{E}[\cos(W_t - W_s)] = e^{-\frac{1}{2}(t-s)}$$

Step 4: Simplify

Substituting these results back into our main equation yields:

$$\begin{aligned}\mathbb{E}[Y_t | \mathcal{F}_s] &= e^{t/2} \left[\cos W_s \cdot (0) + \sin W_s \cdot e^{-\frac{1}{2}(t-s)} \right] \\ &= e^{t/2} \cdot e^{-t/2+s/2} \sin W_s \\ &= e^{s/2} \sin W_s \\ &= Y_s\end{aligned}$$

Since $\mathbb{E}[Y_t | \mathcal{F}_s] = Y_s$, the process Y_t is a martingale.

Answer: Yes

Question 6:

Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion (GBM) with drift parameter $\mu = 0.01$ and volatility parameter $\sigma = 0.3$. If $Y(0) = 5$. Then, find the value of $E[Y(10)]$.

Answer: $5 e^{0.55}$.

Solution

Expected Value of a Geometric Brownian Motion (GBM)

Depending on the definition of the drift parameter μ used in your framework, there are two standard ways to calculate $E[Y(10)]$.

Method 1: Standard SDE Interpretation

If the process is defined via the Stochastic Differential Equation (SDE):

$$dY(t) = \mu Y(t)dt + \sigma Y(t)dW(t)$$

The solution to the SDE is $Y(t) = Y(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right)$. The expected value depends only on the drift coefficient μ :

$$E[Y(t)] = Y(0)e^{\mu t}$$

Given parameters: $Y(0) = 5$, $\mu = 0.01$, $\sigma = 0.3$, and $t = 10$:

$$E[Y(10)] = 5 \cdot e^{0.01 \cdot 10} = 5 \cdot e^{0.1} \approx \mathbf{5.5259}$$

Method 2: Log-Normal / Exponential Interpretation

If the process is defined directly by the exponent equation:

$$Y(t) = Y(0)e^{\mu t + \sigma W(t)}$$

where μ is the drift of the underlying Brownian motion, the expectation includes the volatility adjustment due to Itô's lemma:

$$E[Y(t)] = Y(0)e^{(\mu + \frac{1}{2}\sigma^2)t}$$

Substituting the given parameters:

$$\mu + \frac{1}{2}\sigma^2 = 0.01 + \frac{1}{2}(0.3)^2 = 0.01 + 0.045 = 0.055$$

$$E[Y(10)] = 5 \cdot e^{0.055 \cdot 10} = 5 \cdot e^{0.55} \approx \mathbf{8.6663}$$

Alter We have using the formula in class

$$\begin{aligned} E[Y(10)] &= 5 \cdot e^{(0.01 + (0.3)^2/2) \cdot 10} \\ &= 5 \cdot e^{0.55} \end{aligned}$$

Thus, the expected value is $E[Y(10)] = 5 \cdot e^{0.55}$

Answer: $5 \cdot e^{0.55}$

Question 7:

Weiner process is a Gaussian process. (T/F)

Answer: T

Question 8:

Which of the following statements is TRUE?

- (a) Every stochastic process is a martingale.
- (b) The mean of a martingale process is time varying.
- (c) The mean of a martingale process is constant with respect to time.
- (d) The mean of a martingale process need not exist.

Answer: (c) The mean of a martingale process is constant with respect to time.

Question9 :

Let $Y_t := e^{W_t}$. Then $Cov(Y_1, Y_2) =$

Answer: $e^{3/2}(e - 1)$

Solution Let $Y_t := e^{W_t}$, where W_t is a standard Brownian motion. We want to find $Cov(Y_1, Y_2)$.

The covariance formula is given by:

$$Cov(Y_1, Y_2) = E[Y_1 Y_2] - E[Y_1]E[Y_2]$$

1. Compute $E[Y_1]$ and $E[Y_2]$

Since $W_t \sim N(0, t)$, we use the log-normal property $E[e^{W_t}] = e^{\frac{1}{2}t}$:

- $E[Y_1] = E[e^{W_1}] = e^{1/2}$
- $E[Y_2] = E[e^{W_2}] = e^{2/2} = e^1$

Multiplying them together yields:

$$E[Y_1]E[Y_2] = e^{1/2} \cdot e^1 = e^{3/2}$$

2. Compute $E[Y_1 Y_2]$

We rewrite the product using exponents:

$$E[Y_1 Y_2] = E[e^{W_1} e^{W_2}] = E[e^{W_1 + W_2}]$$

The random variables W_1 and W_2 are jointly normal, so their sum $W_1 + W_2$ is normally distributed with:

- **Mean:** $E[W_1 + W_2] = 0$
- **Variance:**

$$\begin{aligned}\text{Var}(W_1 + W_2) &= \text{Var}(W_1) + \text{Var}(W_2) + 2 \text{Cov}(W_1, W_2) \\ &= 1 + 2 + 2 \min(1, 2) \\ &= 1 + 2 + 2(1) = 5\end{aligned}$$

Thus, $(W_1 + W_2) \sim N(0, 5)$, which gives:

$$E[e^{W_1 + W_2}] = e^{0 + \frac{1}{2}(5)} = e^{5/2}$$

3. Final Result

Subtracting the product of the expectations from the joint expectation:

$$\text{Cov}(Y_1, Y_2) = e^{5/2} - e^{3/2} = e^{3/2}(e - 1)$$

Answer:

Question 10 :

A person is playing a game in which he/she wins or loses one rupee in each round of play, with equal chances and independently of the past outcomes. If he/she starts playing with the firm determination that he/she will stop playing when either he/she won 10 rupees or she lost 5 rupees.

- (a) What is the probability that he/she will be winning when he/she stops playing further.
- (b) What is the expected number of his/her playing rounds before he/she will stop playing further.

Answer: (a) $1/3$, (b) 50

Solution

Solution using Gambler's Ruin Problem

This problem can be modeled as a classic **Gambler's Ruin** problem (a random walk with two absorbing barriers).

Let us shift the scale so that the player's initial position is z , the lower absorbing boundary (losing 5 rupees) is 0, and the upper absorbing boundary (winning 10 rupees) is a .

- Lower boundary: 0
- Initial capital: $z = 5$
- Upper boundary: $a = 5 + 10 = 15$
- Probability of winning a round: $p = 0.5$
- Probability of losing a round: $q = 1 - p = 0.5$

(a) Probability that he/she will be winning when he/she stops playing

For a fair game where $p = q = 0.5$, the probability P_z of reaching the upper barrier a starting from an initial position z is given by the formula:

$$P_z = \frac{z}{a}$$

Substituting the given values:

$$P_5 = \frac{5}{15} = \frac{1}{3} \approx 0.3333$$

Therefore, the probability that the player stops as a winner is $\frac{1}{3}$ (or **33.33%**).

(b) Expected number of playing rounds before he/she stops playing

For a fair game where $p = q = 0.5$, the expected number of rounds E_z to reach either of the two boundaries starting from z is given by the formula:

$$E_z = z(a - z)$$

Substituting the given values:

$$E_5 = 5 \times (15 - 5) = 5 \times 10 = 50$$

Therefore, the expected number of playing rounds before stopping is **50 rounds**.

Alternative solution Solution Using Martingales

Let X_n be the net winnings of the player after n rounds. We can express this as:

$$X_n = \sum_{i=1}^n Y_i$$

where Y_i are independent and identically distributed (i.i.d.) random variables representing each round's outcome, with:

$$P(Y_i = 1) = \frac{1}{2}, \quad P(Y_i = -1) = \frac{1}{2}$$

This discrete-time symmetric random walk has two well-known martingales (analogous to the continuous-time Brownian motion martingales W_t and $W_t^2 - t$):

1. $M_n^{(1)} = X_n$ is a martingale.
2. $M_n^{(2)} = X_n^2 - n$ is a martingale.

Let τ be the stopping time when the player stops playing:

$$\tau = \min\{n \geq 0 : X_n = 10 \text{ or } X_n = -5\}$$

By the **Optional Stopping Theorem (OST)**, since the stopping time τ is almost surely finite and the process is bounded up to τ , the expected value of these martingales at τ equals their initial values at $n = 0$.

(a) Probability of Winning

Using the first martingale, $M_n^{(1)} = X_n$, with initial value $X_0 = 0$:

$$E[X_\tau] = E[X_0] = 0$$

Let p be the probability that the player wins (reaches +10). The probability of losing (reaching -5) is therefore $1 - p$. Expanding the expectation:

$$\begin{aligned} (10) \cdot p + (-5) \cdot (1 - p) &= 0 \\ 10p - 5 + 5p &= 0 \\ 15p &= 5 \\ p &= \frac{1}{3} \end{aligned}$$

The probability that the player will be winning when they stop playing is $\frac{1}{3}$.

(b) Expected Number of Playing Rounds

Using the second martingale, $M_n^{(2)} = X_n^2 - n$ (the discrete analogue to $W_t^2 - t$), with initial value $X_0^2 - 0 = 0$:

$$E[X_\tau^2 - \tau] = E[X_0^2 - 0] = 0$$

By the linearity of expectation, we can separate the terms:

$$E[X_\tau^2] - E[\tau] = 0 \implies E[\tau] = E[X_\tau^2]$$

We calculate $E[X_\tau^2]$ by substituting the winning probability $p = \frac{1}{3}$ found in part (a):

$$\begin{aligned} E[\tau] &= (10)^2 \cdot p + (-5)^2 \cdot (1 - p) \\ &= 100 \cdot \left(\frac{1}{3}\right) + 25 \cdot \left(1 - \frac{1}{3}\right) \\ &= \frac{100}{3} + 25 \cdot \left(\frac{2}{3}\right) \\ &= \frac{100}{3} + \frac{50}{3} \\ &= \frac{150}{3} = \mathbf{50} \end{aligned}$$

The expected number of playing rounds before the player stops playing is **50**.

Answer: (a) $1/3$, (b) 50